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一 研究方向

生态遥感、遥感信息解译、遥感建模技术、影像数据处理

二 教育背景

理学博士 **北京师范大学 地理科学学部** **2021.09 – 2025.06**
专业：地图学与地理信息系统 (**A+**学科) 导师：潘耀忠教授

工学硕士 **中国矿业大学（北京） 地球科学与测绘工程学院** **2018.09 – 2021.06**
专业：大地测量学与测量工程 (**A-**学科) 导师：阎跃观教授、郭伟副教授

工学学士 **河南理工大学 测绘与国土信息工程学院** **2014.09 – 2018.06**
专业：测绘工程

三 工作经历

博士后 **杭州市北京航空航天大学国际创新研究院（北京航空航天大学国际创新学院）** **2025.07 – 至今**

四 科研成果

1 第一/通讯作者论文

[1] **Chuanwu Zhao**, Huijie Zhao, Yaozhong Pan, Guifei Jing. Optimizing Spectral Inputs for Deep Learning-Based Vegetation Destruction Detection Using Sentinel-2 Imagery. *Remote Sensing*. 2026, 18(14), 2333. (SCI, 影响因子 4.3)

[2] **Chuanwu Zhao**, Yaozhong Pan, Hanyi Wu, Yu Zhu. A new effective and robust index using spectral curve shapes and local reflectance peaks: detecting simple coated steel-tiled houses in urban areas. *IEEE Transactions on Geoscience and Remote Sensing*, 2025, 63, 4404716. (SCI, 影响因子 9.4, 引用 2 次)

[3] **Chuanwu Zhao**, Haishuo Wei, Gudina Legese Feyisa, Thiaggo de Castro Tayer, Gelilan Ma, Hanyi Wu, Yaozhong Pan. Evaluating spectral indices for water extraction: limitations and contextual usage recommendations. *International Journal of Applied Earth Observation and Geoinformation*, 2025, 139, 104510. (SCI, 影响因子 8.2, 引用 33 次)

[4] **Chuanwu Zhao**, Yaozhong Pan, Shoujia Ren, Yuan Gao, Hanyi Wu, Gelilan Ma. Accurate vegetation destruction detection using remote sensing imagery based on the three-band difference vegetation index (TBDVI) and dual-temporal detection method. *International Journal of Applied Earth Observation and Geoinformation*, 2024, 127, 103669. (SCI, 影响因子 8.2, 引用 35 次)

[5] **Chuanwu Zhao**, Yaozhong Pan, Hanyi Wu, Shoujia Ren, Gelilan Ma, Yuan Gao, Yu Zhu, Guifei Jing. A novel spectral index for vegetation destruction event detection based on multispectral remote sensing imagery. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 2024, 17, 11290-11309. (SCI, 影响因子 6.3, 引用 11 次)

[6] **Chuanwu Zhao**, Yaozhong Pan, Peng Zhang. Development of a new indicator for identifying vegetation destruction events using remote sensing data. *Ecological Indicators*, 2024, 166, 112553. (SCI, 影响因子 8.7, 引用 20 次)

[7] **Chuanwu Zhao**, Yaozhong Pan, Hanyi Wu, Yu Zhu. Quantifying the contribution of industrial zones to urban heat islands: Relevance and direct impact. *Environmental Research*, 2024, 240, 117594. (SCI, 影响因子 8.2, 引用 18 次)

[8] **Chuanwu Zhao**, Yaozhong Pan. A novel spectral index for mapping blue colour-coated steel roofs (BCCSRs) in urban areas using Sentinel-2 data. *International Journal of Digital Earth*, 2023, 16(1), 2862-2884. (SCI, 影响因子 5.4, 引用 11 次)

[9] **Chuanwu Zhao**, Yaozhong Pan, Xiufang Zhu, Le Li, Xingsheng Xia, Shoujia Ren, Yuan Gao. Monitoring of deforestation events in the tropics using multidimensional features of Sentinel 1 radar data. *Frontiers in Forests and Global Change*, 2023, 6, 1257806. (SCI, 影响因子 4.3, 引用 14 次)

[10] **Chuanwu Zhao**, Yaozhong Pan, Yongjia Teng, Muhammad Fahad Baqa, Wei Guo. Air quality improvement in China: evidence from PM2.5 concentrations in five urban agglomerations, 2000–2021. *Atmosphere*, 2022, 13(11), 1839. (SCI, 影响因子 2.6, 引用 8 次)

[11] **Chuanwu Zhao**, Yuhuan Zhang, Wei Guo, Muhammad Fahad Baqa. Dynamics and drivers of water clarity derived from Landsat and in-situ measurement data in Hulun lake from 2010 to 2020. *Water*, 2022, 14(8), 1189. (SCI, 影响因子 3.5, 引用 13 次)

[12] Wei Guo, **Chuanwu Zhao#**, Yuhuan Zhang, Shangqing Gao. Mapping impervious surface distribution and dynamics in an arid/semiarid area-A case study in Ordos, China. *IEEE Access*, 2021, 9, 19659-19673. (SCI, 影响因子 4.2, 引用 12 次)

[13] **赵传武**, 吴子若, 张玉环, 郭伟. 基于多时相遥感的鄂尔多斯市不透水面变化分析. 测绘通报, 2022, 5, 56-61. (北大核心, 引用 10 次)

[14] **赵传武**, 郭伟, 阎跃观, 戴华阳, 张建. 典型资源型城市的植被覆盖变化及驱动力分析——以鄂尔多斯市为例. 遥感技术与应用, 2022, 37(2), 389-398. (北大核心, 引用 13 次)

[15] **赵传武**, 黄宝柱, 阎跃观, 代文晨, 张建. 一种非线性权重的自适应鲸鱼优化算法. 计算机技术与发展, 2020, 30(10), 7-13. (北大核心, 引用 26 次)

2 合作论文

[1] Gelilan Ma, **Chuanwu Zhao**, Hanyi Wu, Yuan Gao, Shoujia Ren, Yaozhong Pan. A novel red-edge-based anomalous water detection index and diagnostic framework. *International Journal of Digital Earth*, 2025, 18(2), 2540078. (SCI, 影响因子 5.4, 引用 13 次)

[2] Hanyi Wu, **Chuanwu Zhao**, Yu Zhu, Yaozhong Pan. A multiscale examination of heat health risk inequality and its drivers in mega-urban agglomeration: A case study in the Yangtze River Delta, China. *Journal of Cleaner Production*, 2024, 458, 142528. (SCI, 影响因子 10.7, 引用 34 次)

[3] Shoujia Ren, Yaozhong Pan, **Chuanwu Zhao**, Yuan Gao, Gelilan Ma. ASAI: A general and training-free artificial surfaces anomaly index using post-disaster single-temporal and high-resolution imagery. International Journal of Applied Earth Observation and Geoinformation, 2026, 146, 105050. (SCI, 影响因子 8.2)

[4] Mingxiu Tang, Xiufang Zhu, **Chuanwu Zhao**, Junying Song, Chang Xiao, Rui Guo. REBAI: Development and validation of a novel indicator for burned area detection using Sentinel-2 images. Ecological Indicators, 2026, 182, 114501. (SCI, 影响因子 8.7, 引用 1 次)

[5] Xianglin Ji, Yilin Sun, Wei Guo, **Chuanwu Zhao**, Kai Li. Land use and habitat quality change in the Yellow River Basin: A perspective with different CMIP6-based scenarios and multiple scales. Journal of Environmental Management, 2023, 345(12), 118729. (SCI, 影响因子 9.2, 引用 86 次)

[6] Ying Cui, Guobin Shi, **Chuanwu Zhao**, Qingqing Li, Hansi Yao, Wei Guo. A global continuous 500 m nighttime light dataset (1992–2024) via NDVI-guided DMSP-OLS correction and U-TransNet cross-sensor harmonization. *International Journal of Applied Earth Observation and Geoinformation*, 2026, 148, 105230. (SCI)

[7] Wei Guo, Yongjia Teng, Yueguan Yan, **Chuanwu Zhao**, Wanqiu Zhang, Xianglin Ji. Simulation of land use and carbon storage evolution in multi-scenario: A case study in Beijing-Tianjin-Hebei urban agglomeration, China. *Sustainability*, 2022, 14(20), 13436. (SCI, 影响因子 4.1, 引用 39 次)

[8] Shoujia Ren, Yaozhong Pan, Xiufang Zhu, **Chuanwu Zhao**, Yuan Gao. A general and simple automated impervious surface mapping approach based on three-dimensional texture features (3DTF) using fine spatial resolution remotely sensed imagery. Science of The Total Environment, 2024, 923, 171181. (SCI, 影响因子 9.8, 引用 6 次)

[9] Yu Zhu, Yaozhong Pan, Dujuan Zhang, Hanyi Wu, **Chuanwu Zhao**. A Deep Learning Method for Cultivated Land Parcels' (CLPs) Delineation From High-Resolution Remote Sensing Images With High-Generalization Capability. IEEE Transactions on Geoscience and Remote Sensing, 2024, 62, 441525. (SCI, 影响因子 9.4, 引用 2 次)

[10] Yu Zhu, Yaozhong Pan, Tangao Hu, Dujuan Zhang, **Chuanwu Zhao**, Yuan Gao. A generalized framework for agricultural field delineation from high-resolution satellite imageries. International Journal of Digital Earth, 2024, 17(1), 2297947. (SCI, 影响因子 5.4, 引用 22 次)

[11] Wei Guo, Yongjia Teng, Jing Li, Yueguan Yan, **Chuanwu Zhao**, Yongxing Li, Xiang Li. A new assessment framework to forecast land use and carbon storage under different SSP-RCP scenarios in China. *Science of The Total Environment*, 2024, 912, 169088. (SCI, 影响因子 9.8, 引用 76 次)

[12] Yuan Gao, Yaozhong Pan, Xiufang Zhu, Le Li, Shoujia Ren, **Chuanwu Zhao**, Xuechang Zheng. FARM: A fully automated rice mapping framework combining Sentinel-1 SAR and Sentinel-2 multi-temporal imagery. *Computers and Electronics in Agriculture*, 2023, 21, 108262. (SCI, 影响因子 10.3, 引用 56 次)

[13] Yuan Gao, Yaozhong Pan, Xiufang Zhu, Hanyi Wu, Shoujia Ren, **Chuanwu Zhao**. NDCI-mGMM: a novel and automated model for dynamic maize mapping during the growing season. *GIScience & Remote Sensing*, 2026, 63(1), 2606468. (SCI, 影响因子 6.8, 引用 3 次)

[14] Lingang Wang, Yuan Gao, Yaozhong Pan, Wenjie Feng, Xingsheng Xia, **Chuanwu Zhao**, Xuechang Zheng, Hongyan Che, Yongsheng Huang, Xuehui Hou. A novel spectral and phenological composite index for the early automatic mapping of potatoes from Sentinel-2 multi-temporal images. *Computers and Electronics in Agriculture*, 2026, 244, 111459. (SCI, 影响因子 10.3)

[15] Bang Zhou, Yueguan Yan, Ming Li, Shengcai Li, **Chuanwu Zhao**, Jianrong Kang, Jinman Zhang. Integrating Loose Layer Drainage into Mining Subsidence Prediction: A Mathematical Model Validated by Field Measurements and Numerical Simulations. Water, 2025, 17(18), 2687. (SCI, 影响因子 3.5)

[16] 季翔林, 阎跃观, 郭伟, 滕永佳, **赵传武**. 耦合遥感生态指数模型的山西省及规划矿区生态环境评价. 煤田地质与勘探, 2023, 51(3), 103-112. (北大核心)

[17] 滕永佳, 阎跃观, 郭伟, 李凯, **赵传武**. 基于遥感生态指数的矿业城市生态环境时空变异分析. 金属矿山, 2023, 9, 213-222. (北大核心)

[18] 代安进, 戴华阳, 阎跃观, 朱元昊, **赵传武**. 基于 Delaunay 三角网约束的倾斜影像匹配. 湖南科技大学学报(自然科学版), 2023, 38(2). 35-40. (北大核心)

[19] 郭伟, 杨春宇, 吴子若, 季翔林, 杨春洁, **赵传武**, 张玉环. 无人机高分辨率影像的森林株数密度制图. 测绘通报, 2021, 7, 29-33. (北大核心)

[20] 袁德宝, 张建, **赵传武**, 杜世高, 彭金英. 基于改进 RBF 神经网络的 GNSS 高程拟合. 大地测量与地球动力学, 2020, 40(3), 221-224. (北大核心)

3 会议论文

[1] **Chuanwu Zhao**, Yaozhong Pan#, Shoujia Ren, Gelilan Ma, Yuan Gao, Hanyi Wu, Yu Zhu. Improving the detection accuracy of vegetation destruction events using bands sensitive to vegetation foliage, canopy and water content. EGU General Assembly 2024, Vienna, Austria, EGU24-1217, https://doi.org/10.5194/egusphere-egu24-1217. (Abstract)

[2] **Chuanwu Zhao**, Yaozhong Pan#, Shoujia Ren, Yuan Gao, Gelilan Ma, Hanyi Wu. Identifying small and scattered red roofs in urban areas using multispectral remote sensing imagery. IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece.

[3] Shoujia Ren, Yaozhong Pan#, **Chuanwu Zhao**, Yuan Gao, Gelilan Ma, Hanyi Wu, Yu Zhu, Zhengyang Zhang. A novel artificial surface anomaly index (ASAI) based on post-disaster texture features using single-temporal and high-resolution imagery, EGU General Assembly 2024, Vienna, Austria, EGU24-1228, https://doi.org/10.5194/egusphere-egu24-1228. (Abstract)

[4] Hanyi Wu, Yaozhong Pan#, Yuan Gao, Decai Jin, **Chuanwu Zhao**, Shoujia Ren. Improving sample applicability for early-season mapping of winter wheat using geoclimatic zoning. IGARSS 2024 - 2024 IEEE International Geoscience and Remote Sensing Symposium, Athens, Greece. (EI)

[5] Yuan Gao, Yaozhong Pan #, Shoujia Ren, **Chuanwu Zhao**. Comparison of multi-texture features extracts from GF-1 images to map agricultural plastic greenhouse through an object-based approach. IGARSS 2023 - 2023 IEEE International Geoscience and Remote Sensing Symposium, California, United States. (EI)

(# 代表通讯作者)

4 发明专利

[1] 潘耀忠, **赵传武**, 任首佳, 郑学昌, 齐文栋. 一种森林破坏区识别方法、系统、电子设备及存储介质. 中国专利, 2025, ZL202310369038.5.

[2] 潘耀忠, 任首佳, **赵传武**, 郑学昌, 高媛, 齐文栋. 一种不透水面制图方法、系统、设备及介质. 中国专利, 2025, ZL202310708256.7.

[3] 王林刚, 夏兴生, 潘耀忠, 高媛, **赵传武**. 基于遥感指数的马铃薯早期制图方法、装置、介质及产品. 中国专利, 2024, CN202410772366.4.

[4] 潘耀忠, 吴瀚逸, 朱秀芳, 靳德才, 高媛, **赵传武**, 代佳佳. 一种农作物全生长季多光谱遥感时间序列影像近实时重建方法、装置、设备及介质. 中国专利, 2025, CN202411517750.6.

五 科研项目

[1] 国家自然科学基金重大项目, 地表异常遥感探测与即时诊断方法. 2022.1-2026.12. (参与)

[2] 国家重点研发计划, 灾害风险要素多模态遥感特性和建模分析技术. 2022.1-2025.12. (参与)

[3] 国家重点研发计划, 全球粮食与病虫害监测与预警. 2024.1-2026.12. (参与)

[4] 省级横向课题, 北京市生态环境遥感监测. 2022.1-2025.12. (参与)

[5] 遥感与数字地球全国重点实验室-北京市陆表遥感数据产品工程技术研究中心 2024 年开放课题, 基于 Sentinel-2 影像的复杂城市场景涂层钢瓦提取研究. 2024.7-2025.6. (主持)

[6] 遥感与数字地球全国重点实验室-北京市陆表遥感数据产品工程技术研究中心 2022 年开放课题, 基于 Sentinel-1 数据的森林砍伐检测方法研究. 2022.7-2024.1. (主持)

[7] 中国博士后科学基金管理委员会第 78 批面上资助, 突发性植被破坏事件的光学与雷达遥感协同识别方法及典型案例应用. 2026.1-2027.12. (主持)

六 奖励荣誉

2026	研究生学术创新奖“一等奖” 北京师范大学
2025	研究生学术创新奖“特等奖” 北京师范大学
2025	北京市优秀毕业生
2025	校级优秀毕业生 北京师范大学
2023	校三好学生 北京师范大学
2023 & 2022	博士一等奖学金 北京师范大学
2021	新生奖学金 北京师范大学
2021	北京市优秀毕业生
2021	校级优秀毕业生 中国矿业大学（北京）
2020 & 2019	校级研究生优秀学生 中国矿业大学（北京）
2020	研究生二等奖学金 中国矿业大学（北京）
2019	研究生一等奖学金 中国矿业大学（北京）
2018	新生奖学金 中国矿业大学（北京）

七 科研活动

1 学术会议

- 2026: 第一届即时遥感国际研讨会，北京
- 2025: 第二届浙江省遥感科学前沿交叉论坛，浙江杭州
- 2025: 第二十三届中国遥感大会，广东广州
- 2024: 2024 年欧洲地球科学联盟（EGU）年会，奥地利维也纳
- 2024: 第 44 届国际地球科学与遥感研讨会(IGRASS)，希腊雅典
- 2023: 第 43 届国际地球科学与遥感研讨会(IGRASS)，美国洛杉矶
- 2023: 第六届全国定量遥感学术论坛，四川成都

2 学术服务与期刊工作

- IEEE 会员、浙江省地理学会会员、浙江省遥感学会会员、《测绘工程》期刊青年梯队委员;
- 担任 Remote Sensing 的“Vegetation Dynamics Monitoring Using Satellite Remote Sensing”专栏编辑;
- 担任 IEEE Transactions on Geoscience and Remote Sensing, IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, GIScience & Remote Sensing, International Journal of Digital Earth, Geo-spatial Information Science, Ecology and Evolution, Remote Sensing, IEEE ACCESS, Land, Sustainability, PLOS ONE 等期刊审稿人.